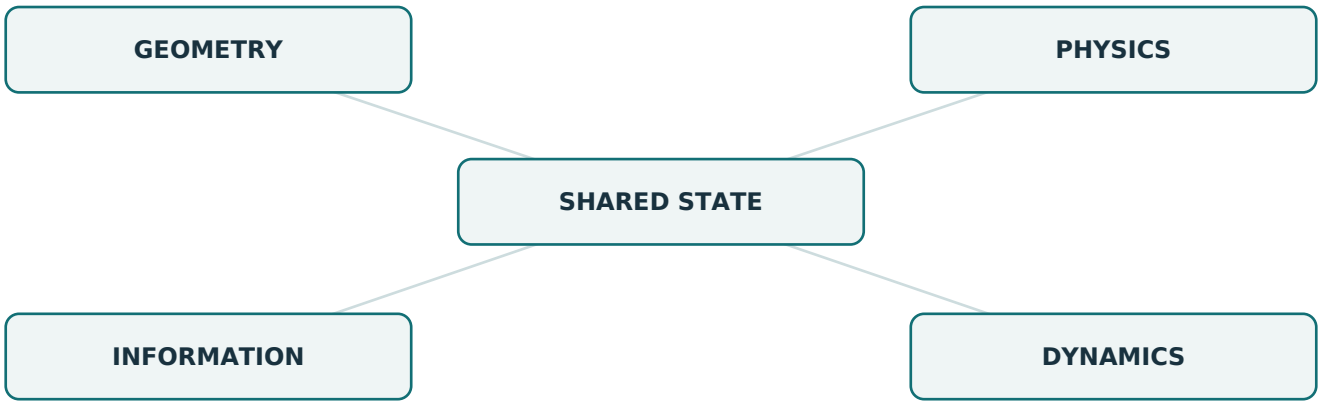
XD-TAU

An Integrated Architecture of Geometry, Physics and Information

Binary-spherical organization.
Continuous fields. Coupled physical domains.

A formal expansion of the Binary-Spherical Organogram Architecture through constructive geometry, material dynamics, transport, control, graphs, coding and spacetime.



EXPANDED DOMAINS / FORMAL CONNECTIONS / DIRECT EVOLUTION

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The Diff-Force-CSG expansion, with mathematical domains, physical units and explicit inter-domain mappings.

SYNTHESIS / 02

The expanded domain map

XD-TAU coordinates addressable entities, implicit geometry and evolving fields through shared state. The names below identify formal representations or specialized adapters, each defined by equations in this edition.

Domain	Formal connection	Pages
Geometry / CAD	Functional representation; regularized CSG; differentiable parameter fitting	4-6
Topology	Klein immersion, thickened image and level-set evolution	7
Geospatial computing	Binary equal-area hierarchy, zonal lookup and spherical metrics	8-9
Information visualization	Log-polar hierarchy and Poincaré focus/context geometry	10
Analog coordinate design	Shark profile, geyser map, barycentric chart and spiral phase	11
Continuum mechanics	Control volumes, material deformation, stress and traction	12-13
Fluid / thermal physics	Mass, momentum, energy, plume transport and drag-to-heat transfer	14-16
Optics	Absorption, radiative source coupling and material response	16
Control / cybernetics	Port-Hamiltonian coupling and self-referential state evolution	6, 17, 20
Graphs / neuroscience	Laplacian diffusion, neural fields and connectivity signals	18
Hardware / data structures	Packed topology, rank/select, XOR kernels and propagation time	19
Distributed systems	Join-semilattice operation logs and replicated state	20
Coding / quantum information	Linear parity, graph-state stabilizers and nonlinear Boolean channels	21-22
Spacetime geometry	3+1 metric, lapse, shift and ADM state	23
Orbital / network systems	Ephemerides, directional cells, control and beamforming	24

Pages 25-26 connect the domains through coupled profiles, reaction networks and scale-patterned thermodynamics. Pages 27-28 record 27 formula checks; pages 29-30 give the references and edition lineage.

FOUNDATION / 03

State, time and dimensional meaning

$$s_n = (\mathsf{E}_n, \mathsf{H}_n, P_n, \xi_n, b_n, \mathsf{F}_n, \mathsf{O}_n, \mathcal{C}_n) \tag{1.1}$$

E is the entity table; H the address hierarchy; P the parenthesized geometry program; ξ the continuous parameters; b the packed bits; F the physical and graph fields; O the tagged operation set; and C the declared couplings. Each module reads the current state and contributes to the next state.

$$e = (\mathrm{ID}, \mathrm{orgParent}, w, \mathbf{x}, \mathrm{payload}) \tag{1.2}$$

An ID identifies an entity. The binary word w identifies a spatial region. Organizational parents, spatial coordinates and interaction edges are distinct stored relations. A subtree query resolves entity IDs before resolving their distributed positions.

One clock, several time coordinates

$$\tau_\circ = 2\pi, \quad n \in \mathbb{N}, \quad T = \ln(1 + t/t_*), \quad t_* > 0,\ t \geq 0 \tag{1.3}$$

$$t = t_*(e^T - 1), \qquad \partial_T = (t + t_*)\partial_t \tag{1.4}$$

The full turn τ_circle is angular. n is the update index. t is physical time, and T is its dimensionless logarithmic coordinate. Physical increments follow:

$$\Delta t_n = t_*(e^{T_{n+1}} - e^{T_n}) \tag{1.5}$$

Quantity	Domain and units
Position, radius, exact distance	Euclidean R^3 or a declared manifold; metres
Field Φ and blend width h	Compatible scalar-field units; length for distance-valued primitives
Velocity, force, energy	m/s, N and J in a physical adapter
Density, pressure, temperature	kg/m³, Pa and K
Optimization state ξ and loss L	Nondimensional coordinates with stored parameter scales
Bits, masks and adjacency	Finite binary vectors and matrices over F_2

Domain definitions specify the inputs for each equation. Couplings record units, coordinate frames, sample locations and time epochs, so a geometric parameter, an information bit and a physical force keep their respective meanings.

GEOMETRY / CAD / 04

Functional representation and full CSG

A primitive supplies a continuous defining field a(x), negative in its solid interior. A geometry program combines primitive fields through a parenthesized expression tree. This is the functional-representation connection [R1].

$$\Omega(\Phi) = \overline{\{\mathbf{x} : \Phi(\mathbf{x}) < 0\}} \tag{2.1}$$

$$\Phi_{A \cup B} = \min(a, b), \qquad \Phi_{A \cap B} = \max(a, b) \tag{2.2}$$

$$\Phi_{A \setminus B} = \max(a, -b) \tag{2.3}$$

These define regularized union, intersection and difference. The strict interior followed by closure removes lower-dimensional remnants: subtracting a solid from itself gives |a|, whose negative interior is empty.

Occupancy and total distance semantics

$$q_\Omega(\mathbf{x}) = \mathbf{1}_\Omega(\mathbf{x}) \tag{2.4}$$

$$d_\Omega(\mathbf{x}) = \operatorname{dist}(\mathbf{x}, \Omega) - \operatorname{dist}(\mathbf{x}, \mathbb{R}^3 \setminus \Omega) \tag{2.5}$$

$$\operatorname{dist}(\mathbf{x}, \varnothing) = +\infty, \quad d_\varnothing = +\infty, \quad d_{\mathbb{R}^3} = -\infty \tag{2.6}$$

Empty, bounded, unbounded and full-space regions are ordinary geometric outcomes. Proper closed regions have the usual negative-inside signed distance. Empty occupancy is zero; full-space occupancy is one. Finite implicit fields remain the channel used for differentiation.

Field types

Each output carries a type: **membership field**, **distance bound**, or **exact signed distance**. Boolean composition produces an implicit region; the distance operator in (2.5) derives its metric field. A sign-modulated Klein component can therefore generate any of the geometric outcomes above through direct evaluation.

Hard min/max compositions of Lipschitz fields are continuous and differentiable almost everywhere. Branch ties identify the locations where a branch derivative changes.

DIFFERENTIABLE GEOMETRY / 05

Smooth CSG and analytic derivatives

The original 45-degree encircled blend becomes the smooth-min operator S_h. Its dual T_h supplies smooth intersection and subtraction. Let a, b and h carry the same field units, with h > 0.

$$D_h = \sqrt{(a-b)^2 + 2h^2} \tag{3.1}$$

$$S_h(a,b) = \tfrac{a+b-D_h}{2}, \qquad T_h(a,b) = \tfrac{a+b+D_h}{2} \tag{3.2}$$

$$\Phi_{\text{union}} = S_h(a,b), \quad \Phi_{\text{intersection}} = T_h(a,b), \quad \Phi_{\text{difference}} = T_h(a,-b) \tag{3.3}$$

$$\partial_a S_h = \tfrac{1}{2}\big(1 - \tfrac{a-b}{D_h}\big), \quad \partial_b S_h = \tfrac{1}{2}\big(1 + \tfrac{a-b}{D_h}\big) \tag{3.4}$$

$$\partial_h S_h = -h/D_h, \qquad \partial_h T_h = h/D_h \tag{3.4b}$$

$$0 < \min(a,b) - S_h(a,b) \le h/\sqrt{2} \tag{3.5}$$

Equation (3.5) bounds the scalar-field change. The geometry tree retains its parentheses because S_h is nonassociative. Chain rule propagation gives derivatives with respect to coordinates, primitive dimensions and transforms.

Normals and learned leaves

$$\mathbf{n} = \tfrac{\nabla\Phi}{\|\nabla\Phi\|}, \qquad \Phi = 0, \quad \nabla\Phi \neq 0 \tag{3.6}$$

$$f_\theta(\mathbf{x}) = f_{\text{analytic}}(\mathbf{x}) + r_\theta(\mathbf{x}) \tag{3.7}$$

A learned residual r_θ or a neural implicit leaf supplies the neural-geometry specialization. The leaf stores network parameters, input normalization and a fitting objective. Analytical and learned leaves share the same constructive tree [R3].

An image-fitting adapter may use differentiable rasterization, including the primitive-intersection gradients described by DiffCSG [R2]. Field fitting uses the analytic derivatives above.

DYNAMICS / CYBERNETICS / 06

Optimization force and direct evolution

Let ξ be nondimensional continuous parameters, P the geometry expression and b the discrete state. Store dimensional parameters as $\theta = \theta_{\text{scale}} \odot \xi$. A fitting objective defines an optimization direction.

$$\hat{q}_\xi(\mathbf{x}) = \sigma(-\Phi_{\xi,P}(\mathbf{x})/\varepsilon), \quad \sigma(z) = (1+e^{-z})^{-1}, \quad \varepsilon > 0 \tag{4.1}$$

$$L(\xi) = \tfrac{1}{N}\sum_i (\hat{q}_\xi(\mathbf{x}_i) - y_i)^2, \qquad \xi_{n+1} = \xi_n - \eta G_\xi^{-1} \nabla_\xi L \tag{4.2}$$

G_ξ is a positive-definite parameter metric and η is a step size. The vector $-G_\xi^{-1}\nabla L$ is the optimization “force.” Mechanical forces enter the physical momentum equation in newtons.

Continuous, discrete and topology channels

$$b_{n+1} = b_n \oplus u_n, \qquad P_{n+1} = \mathrm{Edit}(P_n, e_n) \tag{4.3}$$

$$s_{n+1} = \mathcal{U}(s_n, e_n), \qquad B_{n+1} = \mathrm{Encode}(s_{n+1}) \tag{4.4}$$

The update reads one state snapshot. Continuous fitting, bit mutations, physical field steps, entity events and expression edits produce the next state directly. Each coupled profile declares its execution order or simultaneous evaluation rule. Occupancy samples and continuous field derivatives are available together.

Local automata and self-reference

$$b_i^{n+1} = u_\theta(\{b_j^n : j \in \mathcal{N}(i)\}, q_i^n), \qquad \theta^{n+1} = v(\theta^n, b^n, q^n) \tag{4.5}$$

Uniform local rules give a graph cellular automaton. Including θ and the geometry program in the encoded state gives a self-referential geometric state machine. A boundary-regenerating module is specified by an explicit production rule that rebuilds its defining region from its own products; this supplies the constructive autopoietic specialization.

Coupling C maps declared outputs into other modules’ inputs. Display deformations, geometry and physical fields may participate in feedback through those explicit couplings.

CONTINUOUS GEOMETRY / 07

Topology, thickened images and interfaces

Let K be the abstract Klein bottle and ι:K→R³ a selected immersion. Its image S is a spatial set. Thickness ε gives a volumetric representation while preserving the distinction between abstract topology and the image geometry.

$$S = \iota(K), \qquad \Omega_{K,\varepsilon} = \{\mathbf{x} : \mathrm{dist}(\mathbf{x}, S) \le \varepsilon\}, \quad \varepsilon > 0 \tag{5.1}$$

$$f_\varepsilon(\mathbf{x}) = \mathrm{dist}(\mathbf{x}, S) - \varepsilon, \qquad d_K = d_{\Omega_{K,\varepsilon}} \tag{5.2}$$

f_ε is an offset membership field. d_K is the derived signed distance. Self-intersections become overlaps in the image's thickened volume. The program stores the immersion, scale and thickness as reproducible parameters.

An evolving interface

$$\partial_t \Phi + \mathbf{v} \cdot \nabla \Phi = 0 \tag{5.3}$$

$$\partial_t \Phi + V_n \|\nabla \Phi\| = 0 \tag{5.4}$$

Advection by velocity v and normal motion with speed V_n provide the Eulerian level-set connection [R4]. V_n can be supplied by curvature, pressure, a fitting direction or a material law. The finite field transports the interface; occupancy and distance are derived outputs.

Geometry, topology and physics

A geometric shell uses a selected solid thickness. A physical shell additionally carries density, stiffness, temperature and material motion. The mechanical adapter maps these fields to stress and traction. Geometric connectivity can change through CSG events; physical interface evolution follows the chosen field equations.

$$\Phi_n = S_h(\min(f_{\mathrm{pyr}}, f_{\mathrm{cone}}), \min(f_{\mathrm{sphere}}, m_n f_K)), \quad m_n(\mathbf{x}) = M_{i_n(\mathbf{x}), j_n(\mathbf{x})} \tag{5.5}$$

Equation (5.5) retains XD-TAU's original accumulated primitive field, now inside the full constructive geometry grammar. The mask supplies piecewise sign modulation; region, occupancy and distance follow (2.1)-(2.6).

Formal connection

The Klein component is an immersed-surface image with a volumetric offset. The level-set component is an evolving implicit interface. Together they provide topology-aware geometric state, with material behavior supplied by the mechanics and transport adapters.

GEOSPATIAL DOMAIN / 08

A concrete binary global grid

Use u = y/R and longitude ψ about the y-axis. The spherical map and its area element are:

$$\mathbf{p}(u,\psi) = R(\sqrt{1-u^2}\cos\psi, u, \sqrt{1-u^2}\sin\psi) \tag{6.1}$$

$$-1 \le u \le 1, \quad 0 \le \psi < 2\pi, \qquad dA = R^2\,du\,d\psi \tag{6.2}$$

Start with $[-1,1]\times[0,2\pi)$. Odd address bits bisect the u interval; even bits bisect the ψ interval. A split boundary belongs to the upper child. The first bit therefore reproduces the original y = 0 hemispherical cut. At either pole, use ψ = 0.

$$n_u = \lceil d/2 \rceil, \quad n_\psi = \lfloor d/2 \rfloor, \quad \mu(C_w) = 4\pi R^2/2^d \tag{6.3}$$

$$j(w) = \sum_{i=1}^{d} b_i 2^{d-i}, \qquad \mathrm{addr}(w) = 2^d - 1 + j(w) \tag{6.4}$$

Every level gives equal-area cells. Midpoint subdivision of u and ψ refines both directions. Pole and seam ownership is unique, with area-zero boundary assignments.

Representatives and metric size

$$\widehat{\mathbf{p}}_w = \mathbf{p}(u_c, \psi_c), \quad u_c = (u_- + u_+)/2, \quad \psi_c = (\psi_- + \psi_+)/2 \tag{6.5}$$

$$\varepsilon_w = R\left[\max_{u\in\{u_-,u_+\}} |\arcsin u - \arcsin u_c| + \tfrac{\psi_+ - \psi_-}{2}\right] \tag{6.6}$$

A meridian-then-parallel path gives d_S(p,p̂_w) ≤ ε_w. Both interval widths decrease with depth, so this radius bound tends to zero. The cell area and radius are metadata for aggregation and resolution selection.

The zonal API supplies locate, cell, representative, parent, children, neighbors and aggregate. Neighbors are cells whose closures share a boundary; cross-seam and polar contacts follow the same definition. This provides the explicit global-grid connection [R5].

SPHERICAL NAVIGATION / 09

OTAN2, coordinate recovery and metrics

The original hemispherical chart remains available beside the equal-area grid. Let h be the y hemisphere, $s_h = 2h-1$ and $r = \sqrt{(x^2+z^2)}$.

$$\rho = \ln(1 + r/r_*), \quad r_* > 0, \qquad 0 \le \rho \le \ln(1 + R/r_*) \tag{7.1}$$

$$q = \mathrm{frac}\left(\frac{\mathrm{atan2}(z, x) + \omega_n}{2\pi}\right), \quad \omega_n = [\omega_0 + \nu T_n]_{2\pi} \tag{7.2}$$

$$r = r_*(e^{\rho} - 1), \qquad \psi = [2\pi q - \omega_n]_{2\pi} \tag{7.3}$$

$$\mathbf{p} = (r\cos\psi, s_h\sqrt{R^2 - r^2}, r\sin\psi) \tag{7.4}$$

The same phase epoch is used for both directions. The seam belongs to h = 1; poles use one designated angular code. The return map satisfies $\|p\|^2 = R^2$.

$$d_S(\mathbf{p}, \mathbf{q}) = R\,\mathrm{atan2}(\|\mathbf{p} \times \mathbf{q}\|, \mathbf{p} \cdot \mathbf{q}) \tag{7.5}$$

Entity records can store exact numerical coordinates or a cell representative plus a residual. Coordinate precision, field sampling resolution and spatial cell size are separate metadata. The regularized radius keeps ρ finite at the poles.

Domain-specific distances

Domain	Metric or geometric object
Sphere navigation	Geodesic distance (7.5) and cell radius ε_w
Solid geometry	Euclidean boundary distance d_Ω
Hyperbolic hierarchy	Poincaré metric on the unit disk
Material mechanics	Reference/deformed metrics and strain
Optimization	Parameter metric G_ξ
Spacetime	Lorentzian line element with lapse and shift

The lookup layer carries the metric type with its data. The pinion supplies angular phase; recorded movement supplies position changes.

INFORMATION VISUALIZATION / 10

Log-polar and hyperbolic organization

A binary hierarchy retains the original L-system production N(w)→N(w0)N(w1). Its log-polar layout separates displayed depth from geographic position.

$$\theta_w = \frac{2\pi(j(w)+1/2)}{2^d}, \quad r_d = r_0 a^d, \quad \rho_d = d\ln a, \quad a > 1 \tag{8.1}$$

Depth is linear in log radius and angular sectors halve at each branch. Entity IDs link this layout to the organizational parent graph and spherical records.

The Poincaré focus/context adapter

$$z_w = \tanh(\kappa d/2)e^{i\theta_w}, \quad \kappa > 0, \qquad ds^2 = \frac{4|dz|^2}{(1-|z|^2)^2} \tag{8.2}$$

$$d_H(0, z_w) = 2\operatorname{artanh}|z_w| = \kappa d \tag{8.3}$$

$$F_a(z) = \frac{z-a}{1-\bar{a}z}, \qquad |a| < 1, \ |z| < 1 \tag{8.4}$$

The Poincaré disk supplies exponential spatial capacity with a finite circular display. F_a moves focus a to the center while preserving hyperbolic distances. This gives the formal focus/context connection [R6].

Hierarchy, interaction and identity

The organizational relation can be nonbinary. Display grouping nodes give a binary layout while original parents remain in the entity records. Interaction graphs can include cycles and weighted edges. Selecting a display node resolves its entity set; moving focus changes the chart coordinates.

A conformal map preserves local angles where its complex derivative is nonzero. The disk automorphism in (8.4) gives a concrete conformal operation. The logarithmic radial layout and the hyperbolic renderer are two explicitly defined display choices.

> **Coupling**
>
> Layout coordinates may drive interaction, sampling or geometric parameters through a declared C map. The selected coupling records which quantities change, while IDs keep the corresponding entities identifiable.

ANALOG GEOMETRY / 11

Shark-Geyser, barycentric and spiral maps

$$f_p(x) = |(x \bmod p) - p/2|, \quad p > 0 \tag{9.1}$$

$$r' = r_0 e^{\alpha_g (r - r_c)}, \quad r_0 > 0, \qquad r = r_c + \alpha_g^{-1} \ln(r'/r_0) \tag{9.2}$$

The triangular profile supplies a periodic folded amplitude. Retaining its period and branch indices gives an inverse description. The geyser map has an inverse for $\alpha_g \neq 0$; α_g carries inverse-length units. A chamber function supplies the accumulation stage.

$$\mathcal{K}(r) = r_c + \ell \tanh((r - r_c)/\ell), \qquad r_c \geq \ell > 0, \quad r \geq 0 \tag{9.2b}$$

This canonical chamber compresses radial increments smoothly; the eruption takes its output as the input radius.

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \mathbf{u} = \sum_i \lambda_i \mathbf{v}_i, \quad \rho_{\text{raw}} = \ln(r/r_*), \ r > 0 \tag{9.3}$$

$$s = \rho/\rho_{\max}, \quad t = \psi/(2\pi), \quad (\lambda_1, \lambda_2, \lambda_3) = (1 - s, s(1 - t), st) \tag{9.3b}$$

$$\text{Len} = s + \gamma_f 2 f_p(s)/p, \qquad \gamma_f \geq 0 \tag{9.3c}$$

The canonical chart uses ρ from (7.1), $\rho_{\max} = \ln(1+R/r_*)$, and a nondegenerate triangle. For $s > 0$, recover $t = \lambda_3/(\lambda_2+\lambda_3)$ and $s = \lambda_2+\lambda_3$. Radial edges identify the seam; $s = 0$ uses the pole code. Nonnegative weights give triangle containment; affine weights also describe its exterior. In Len, p and γ_f are dimensionless.

$$\Theta_{\text{strip}}(\theta) = \theta + \alpha_\theta \tanh((\theta - \beta)/\sigma) \sin(2\pi \text{Len}), \quad \sigma > 0 \tag{9.4}$$

$$\Theta_{\text{circle}}(\theta) = \theta + a \sin(\theta - \beta), \qquad |a| < 1 \tag{9.5}$$

The strip map retains the source's angular profile. The circle map adds a periodic, strictly increasing phase adapter: $\Theta(\theta+2\pi) = \Theta(\theta)+2\pi$ and $\Theta' > 0$. β is a selected axis direction with a stored frame and sign convention.

$$z = k\Theta_{\text{unwrapped}}, \qquad I(r) = \sin(\omega_s \ln(r/r_*)), \quad r > 0 \tag{9.6}$$

$$I(qr) = I(r), \qquad q = e^{2\pi/|\omega_s|}, \quad \omega_s \neq 0 \tag{9.7}$$

The unwrapped phase retains axial displacement across turns. k has length-per-radian units. The log-sine has exact discrete scale repetition. A chosen interval $[r_{\min}, r_{\max}]$ specifies the sampled scales and cycle count for a particular rendering or physical coefficient field.

CONTINUUM DISCRETIZATION / 12

Surface cells, volume cells and flux

Surface quantities live on S^2 or its cell graph. Three-dimensional transport uses radial layers as well as angular cells. Let C_w be a cell on the unit sphere, with solid angle ω_w.

$$K_{w,k} = \{r\mathbf{s} : \mathbf{s} \in C_w, \quad r_k \le r < r_{k+1}\} \tag{10.1}$$

$$|K_{w,k}| = \frac{\omega_w}{3}(r_{k+1}^3 - r_k^3) \tag{10.2}$$

Store face areas, outward normals and neighbor incidence. Shared faces use the same geometric measure and opposite orientations.

$$|K_i|U_i^{n+1} = |K_i|U_i^n - \Delta t \sum_{f \subset \partial K_i} |f|\widehat{F}_{i,f} + \Delta t|K_i|S_i \tag{10.3}$$

U contains cell-averaged conserved variables. F_{hat} is the numerical flux and S is the source. Each internal face contributes equal and opposite transfers to its two cells.

$$\sum_i |K_i|(U_i^{n+1} - U_i^n) = -\Delta t \sum_{f \in \partial \mathcal{D}} |f|\widehat{F}_f + \Delta t \sum_i |K_i|S_i \tag{10.4}$$

Internal cancellation establishes the conservative finite-volume connection. The remaining change is boundary transport plus physical sources. This balance is built into the evolution formula itself.

Surface transport

$$\partial_t c + \nabla_S \cdot (c\mathbf{v}_S) = D\Delta_S c + s \tag{10.5}$$

Here c is an areal density, with tangent velocity v_S, surface divergence and the Laplace-Beltrami operator. This supports population density, ecological transport and spatial signal fields. Surface temperature uses an areal heat-capacity balance. Volume and surface adapters share identities while storing their respective physical measures.

> **Formal connection**
>
> The binary hierarchy supplies addresses and aggregation. Control-volume geometry supplies physical areas and volumes. Antisymmetric shared-face transfers supply conservation.

MECHANICS / 13

Material motion, stress and shells

A material point X moves to x = χ(X,t). The deformation gradient and Green-Lagrange strain define the mechanical meaning of a geometric warp.

$$F = \frac{\partial \chi}{\partial \mathbf{X}}, \quad J = \det F > 0, \quad E_G = \tfrac{1}{2}(F^T F - I) \tag{11.1}$$

$$\rho J = \rho_0, \qquad \rho_0 \ddot{\chi} = \operatorname{Div} P + \rho_0 \mathbf{b} \tag{11.2}$$

The orientation-preserving deformation domain has J > 0. ρ is current density and ρ₀ reference density. P is the first Piola-Kirchhoff stress, with units of force per reference area.

$$P = \frac{\partial W(F, \vartheta)}{\partial F}, \qquad \sigma = J^{-1} P F^T \tag{11.3}$$

W is Helmholtz free-energy density per reference volume, in J/m³; the derivative in (11.3) holds temperature fixed. For small strain ε = (∇u+∇uᵀ)/2, isotropic elasticity gives σ = λ tr(ε)I+2με. Temperature-dependent W supplies thermal expansion and other coupled responses [R11].

Implicit boundaries and pressure

$$\mathbf{n} = \nabla\Phi / \|\nabla\Phi\|, \qquad \mathbf{t} = \sigma \mathbf{n}, \qquad \mathbf{t}_p = -p\mathbf{n} \tag{11.4}$$

Traction acts on regular implicit-surface points. A physical shell uses its thickness, material density, constitutive law and boundary loading. The same cone, sphere or thickened-image geometry can therefore author a mechanical domain.

Coordinate metric and material strain

$$G_\chi = (D\chi)^T D\chi \tag{11.5}$$

G_χ is a pullback metric. When χ maps material points, it measures physical deformation. When χ maps a display chart, it measures chart distortion. The coupling type records which interpretation applies.

The mechanics adapter contributes displacement, velocity, stress, traction and material energy to shared state. Geometry can supply its boundary, and the solved motion can update geometry parameters.

FLUID PHYSICS / 14

Fluid motion and thermodynamic state

Use density ρ, velocity v, pressure p, temperature ϑ and specific internal energy e on a three-dimensional domain. The conserved state is U = (ρ,ρv,ρE), with E = e+|v|²/2.

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{12.1}$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + pI - \tau) = \rho \mathbf{b} \tag{12.2}$$

$$\partial_t (\rho E) + \nabla \cdot [(\rho E + p)\mathbf{v} - \tau \mathbf{v} + \mathbf{q}] = \rho \mathbf{b} \cdot \mathbf{v} + Q \tag{12.3}$$

τ is viscous stress, q heat flux, b body acceleration and Q supplied heat per volume per time. These are the continuum mass, momentum and total-energy balances [R7].

A concrete constitutive family

$$\tau = 2\mu D + \lambda_v (\nabla \cdot \mathbf{v}) I, \quad D = \tfrac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T), \quad \mathbf{q} = -k \nabla \vartheta \tag{12.4}$$

$$p = \rho R_s \vartheta, \qquad e = c_v \vartheta \tag{12.5}$$

The Newtonian ideal-gas family stores shear viscosity μ, second viscosity λ_v, conductivity k, gas constant R_s and specific heat c_v. Bulk viscosity is $\zeta = \lambda_v + 2\mu/3$. Other material models enter through their own stress, heat-flux and equation-of-state functions.

Plumes, convection and entrainment

Gravity and temperature-dependent density produce buoyancy. Velocity transports heat and concentration, pressure redirects the flow, viscosity converts mechanical energy into internal energy, and heat flux redistributes temperature. A plume’s visible envelope is a geometry output of this evolving state.

The finite-volume update on page 12 transports these conserved variables between volume cells. Boundary data and source fields supply inlet flow, ambient conditions, mechanical work and heating. Logical log-time schedules the physical increments through (1.4).

PLUME / ENERGY COUPLING / 15

Shiitake morphology, drag and heat

A cap-and-stem field gives the source's Shiitake morphology. For a toroidal cap and cylindrical stem aligned with z, one useful geometric profile is:

$$f_{\text{cap}} = \sqrt{(\sqrt{x^2+y^2}-a)^2+(z-z_c)^2}-b \tag{13.1}$$

$$f_{\text{stem}} = \max(\sqrt{x^2+y^2}-r_s, |z-z_s|-\ell_s), \quad \Phi_{\text{plume}} = S_h(f_{\text{cap}}, f_{\text{stem}}) \tag{13.2}$$

The cap radius a, tube radius b and stem dimensions are geometric parameters. Density, velocity and temperature fields supply the physical plume state. A selected isovalue or fitted envelope links the two.

Reduced drag channel

$$\mathbf{w} = \mathbf{v}_{\text{body}} - \mathbf{v}_{\text{ambient}}, \qquad \mathbf{F}_D = -\tfrac{1}{2}\rho C_D A \|\mathbf{w}\|\mathbf{w} \tag{13.3}$$

$$P_{\text{diss}} = -\mathbf{F}_D \cdot \mathbf{w} = \tfrac{1}{2}\rho C_D A \|\mathbf{w}\|^3 \tag{13.4}$$

ρ is ambient density, C_D the dimensionless drag coefficient and A the reference area [R8]. Equal and opposite momentum transfers couple body and medium. The dissipated relative mechanical power supplies a heat channel. This reduced model represents unresolved drag.

Energy accounting

$$\frac{d}{dt}(K+U+V) = P_{\text{external}} - \dot{Q}_{\text{outward}} \tag{13.5}$$

$$E_0 = mc^2, \qquad \Delta E_0 = \Delta m\, c^2 \tag{13.6}$$

For a closed-mass body-plus-medium system, K, U and V are kinetic, internal and potential energy. Internal drag transfers K to U; boundary heat and work change the total. Open control volumes use the advective energy flux in (12.3). Equation (13.6) is the rest-energy relation [R10]; a modeled conversion contributes through its explicit source term.

An exact linear-drag example, $m\,\dot{v} = -c_d\,v$ in a closed heat bath, gives $v_{\text{next}} = \exp(-c_d\,\Delta t/m)v$ and heat gain $K - K_{\text{next}}$. The example on page 27 conserves the combined amount directly.

THERMAL / OPTICAL DOMAINS / 16

Diffusion, attenuation and absorption

Thermal and concentration transport

$$\rho c_p(\partial_t \vartheta + \mathbf{v}\cdot\nabla\vartheta) = \nabla\cdot(k\nabla\vartheta) + Q_{\text{heat}} \tag{14.1}$$

$$\partial_t(\rho c) + \nabla\cdot(\rho c\mathbf{v}) = \nabla\cdot(\rho D\nabla c) + S_c \tag{14.2}$$

Equation (14.1) defines a constant-density incompressible thermal adapter with constant specific heat c_p. Compressible transport uses (12.3). In (14.2), c is a dimensionless mass fraction and S_c has units kg/(m³ s). D has units m²/s and k has units W/(m K). Sources include viscous heat, reduced drag or optical absorption.

Amplitude and gyroscopic rotation

$$\dot{A} = -\mu A, \quad A(t) = A_0 e^{-\mu t}, \quad E \propto A^2 \implies E(t) = E_0 e^{-2\mu t} \tag{14.3}$$

$$\mathbf{F}_C = -2m\boldsymbol{\Omega}\times\mathbf{v}, \qquad \mathbf{F}_C\cdot\mathbf{v} = 0 \tag{14.4}$$

μ has units s^{-1}. Amplitude decay and energy decay have their respective rates. Coriolis acceleration redirects velocity while its instantaneous work is zero. Dissipative channels carry their own power terms.

Optical absorption

$$\frac{dI}{ds} = -\kappa I, \quad I(s) = I(0)e^{-\int_0^s \kappa(\xi)d\xi}, \quad Q_{\text{abs}} = \kappa I \tag{14.5}$$

For a collimated absorbing beam, I is irradiance in W/m² and κ an absorption coefficient in m^{-1}. Q_{abs} is deposited heat per volume. Reflectance, transmittance and absorptance satisfy R+T+A = 1 for the declared optical balance. Scattering adds a radiative-transfer operator [R9].

A dark-material domain supplies wavelength-dependent optical coefficients and measured interface response. Geometry controls paths and surface orientation; material fields determine absorption. The source’s material cue becomes this explicit geometry-material-radiation coupling.

PORT-HAMILTONIAN DOMAIN / 17

Energy-based control and interconnection

A common energy language connects mechanics, circuits, transport and control. Let x be the state and H(x) its stored energy. Define a skew operator J, a symmetric dissipative operator R and an input map G [R12].

$$\dot{x} = (J(x) - R(x))\nabla H(x) + G(x)u, \quad J^T = -J, \quad R = R^T \geqslant 0 \tag{15.1}$$

$$y = G^T \nabla H, \qquad \dot{H} = -\nabla H^T R \nabla H + y^T u \tag{15.2}$$

J redistributes energy, R supplies dissipated power, and $y^T u$ is exchanged input power. The operator domains and units determine the state variables: momentum and position, charge and flux, thermal amounts or cell inventories.

Coupling ports

$$u_1 = C_{\text{port}} y_2, \quad u_2 = -C_{\text{port}}^T y_1, \qquad y_1^T u_1 + y_2^T u_2 = 0 \tag{15.3}$$

C_{port} maps compatible power-conjugate units; normalized ports may use the identity. The transfers are equal and opposite in power. Adding subsystem balances cancels their internal exchange. Shared geometry supplies port areas, normals and adjacency.

An explicit mechanical example

$$H(q,p) = \tfrac{p^2}{2m} + \tfrac{kq^2}{2}, \quad \dot{q} = p/m, \quad \dot{p} = -kq - \gamma p/m + u \tag{15.4}$$

$$\dot{H} = -\gamma (p/m)^2 + (p/m)u \tag{15.5}$$

The oscillator’s dissipated power can enter the thermal adapter. This couples macroscopic motion, heat and control through a common state and energy balance.

The architecture’s general coupling C specifies ports and variable transformations. Port-Hamiltonian structure is the energy-based specialization of that map. Its algebra supplies the balance directly through the update model.

GRAPH / NEURAL DOMAINS / 18

Graph signals, neural fields and connectomes

Interpret the symmetric zero-diagonal mask bits as adjacency Q. Add weighted edges W when interaction strength or direction matters. Entity IDs can name people, cells, neurons, sensors or regions.

$$L = D - Q, \quad D_{ii} = \sum_j Q_{ij}, \qquad \dot{x} = -\kappa L x, \quad \kappa > 0 \tag{16.1}$$

$$x^T L x = \sum_{\{i,j\} \in E} (x_i - x_j)^2, \qquad \mathbf{1}^T L = 0 \tag{16.2}$$

This is graph diffusion. On a connected undirected graph it smooths differences and preserves the total signal. Eigenvectors of L give spectral modes for layout, clustering and multiscale signal analysis.

A neural dynamical specialization

$$\dot{x}_i = -\lambda_i x_i + \sum_j W_{ij} \phi(x_j) + u_i \tag{16.3}$$

λ_i is a local decay rate, φ a selected nonlinear response and u_i an external drive. The hierarchy indexes regions; W stores connectivity, including cycles. This defines the computational-connectome interface through actual signals and interactions.

Message-passing geometry

$$h_i^{\ell+1} = \sigma\left(W_0 h_i^{\ell} + \sum_{j \in \mathcal{N}(i)} a_{ij} W_1 h_j^{\ell} \right) \tag{16.4}$$

A learned graph module transforms features on the same entities. Features can include cell coordinates, material coefficients, primitive parameters or observations. A decoder can map h_i to an implicit leaf or a field coefficient; its training loss becomes part of the fitting channel.

For centered entity coordinates $X \in R^{N\times 3}$ and a chosen graph eigenmode $v \in R^N$, define the spatial axis $a = X^T v$. A nonzero xz projection gives $\beta = \operatorname{atan2}(a_z, a_x)$. Store the selected eigenvalue, normalization, sign and reference frame with this spiral-axis adapter.

Shared operator

One graph can support diffusion, spectral visualization, neural interaction and quantum graph-state description. Each adapter assigns its own state and evolution to the same declared adjacency.

DATA STRUCTURES / COMPUTATION / 19

Packed topology, rank/select and hardware

Logical heap addresses retain the original formula 2^d−1+j. A separate breadth-first shape stream D stores two bits per actual node: left-child presence and right-child presence.

$$D[2i+s] = \mathbf{1}[\text{child slot } s \text{ exists}], \quad s \in \{0,1\} \tag{17.1}$$

$$\text{child}(i,s) = \text{rank}_1(D, 2i+s) \quad \text{when } D[2i+s] = 1 \tag{17.2}$$

$$\text{parent}(j) = \lfloor \text{select}_1(D,j)/2 \rfloor, \quad j \geq 1 \tag{17.3}$$

Rank is inclusive at a zero-based position; select uses a one-based occurrence count and returns a zero-based position. Breadth-first node IDs start at zero. This keeps missing-left and missing-right information. The shape uses 2N bits, with an o(N) auxiliary rank/select index in a succinct implementation [R13].

$$\text{bit}(W,i) = (W_{\lfloor i/64 \rfloor} \gg (i \bmod 64)) \text{AND } 1 \tag{17.4}$$

Payloads, coordinates and parameters have their own packed schemas. SIMD/GPU kernels evaluate independent fields or wordwise masks. Node topology, sign masks and entity records retain separate lengths and offsets.

Electrical and logical timing

$$\tau_{RC} = RC, \qquad D_{\text{path}} = \sum_{g \in \text{path}} d_g \tag{17.5}$$

RC is a circuit time constant in seconds; d_g are propagation delays. These quantities connect a physical implementation to clocked or asynchronous scheduling. The full-turn τ_circle and log-time coordinate remain mathematical controls with their stated domains.

Temporal folding is represented concretely by the dependency graph, reuse schedule and measured path delays. The implementation records word width, address representation, memory traffic and kernel timing.

DISTRIBUTED / CYBERNETIC DOMAINS / 20

Replicated state and self-producing rules

A replica stores a set O of uniquely tagged operations. Each tag contains a replica ID and local sequence number; its payload identifies an entity event, parameter assignment or mask mutation.

$$O_a \sqcup O_b = O_a \cup O_b \tag{18.1}$$

$$A \sqcup B = B \sqcup A, \quad (A \sqcup B) \sqcup C = A \sqcup (B \sqcup C), \quad A \sqcup A = A \tag{18.2}$$

Union is associative, commutative and idempotent. Replicas receiving the same tagged operations therefore obtain the same operation set, independent of delivery order and repetition. This is a state-based join-semilattice construction [R14].

$$b(O) = b_0 \oplus \bigoplus_{o \in O} u(o) \tag{18.3}$$

Materializing XOR over distinct operation payloads gives a replicated mask channel. Absolute assignments use a maximum version tuple under a declared total order. Position and organizational-parent assignments retain their respective fields and stable entity IDs.

Self-referential and autopoietic specialization

$$(P_{n+1}, \theta_{n+1}, \Omega_{n+1}) = \mathcal{R}(P_n, \theta_n, \Omega_n, \mathrm{Eval}(P_n, s_n)) \tag{18.4}$$

R is an explicit production rule. It can regenerate its program, parameters and defining boundary from evaluated internal products. A concrete autopoietic module declares the components it produces and the relations by which those components regenerate the module boundary.

Ordinary geometric feedback uses U from page 6. The self-producing specialization adds these regeneration relations. Distributed materialization declares its event ordering for geometry edits and continuous parameter assignments.

> **Formal connection**
>
> Stable IDs support cross-representation identity. The merge algebra supplies replica convergence. The production rule supplies the self-producing cybernetic behavior. These are separately defined operators within shared state.

CODING / QUANTUM INFORMATION / 21

Parity masks and quantum graph states

Retain the double Hadamard product, with elementwise multiplication and sign inputs A,B ∈ {−1,+1}^(N×N).

$$M = (A \odot B) \odot (A^T \odot B^T) = C \odot C^T, \quad C = A \odot B \tag{19.1}$$

$$Q = A_{\text{bit}} + B_{\text{bit}} + A_{\text{bit}}^T + B_{\text{bit}}^T \quad (\mathbb{F}_2) \tag{19.2}$$

The bit/sign convention is $M_{ij} = (-1)^{Q_{ij}}$. M is symmetric with a +1 diagonal. Q is symmetric with a zero diagonal, exactly a simple undirected graph adjacency. The independent mask capacity is N(N−1)/2 bits.

Constructing the stabilizers

$$K_i = X_i \prod_{j\,:\,Q_{ij}=1} Z_j \tag{19.3}$$

$$H = [I \mid Q], \qquad H \Lambda H^T = Q + Q^T = 0 \quad (\mathbb{F}_2) \tag{19.4}$$

Λ is the binary symplectic form exchanging the X and Z halves. Equation (19.4) establishes commutation. The identity block gives N independent generators. Their +1 common eigenspace is the unique graph state for Q [R15].

Coding interpretation

A selected independent commuting subset of r generators defines a stabilizer subspace with N−r logical qubits. The generator phases and chosen subset are explicit quantum-module state. The mask program therefore authors a mathematically defined graph-state description.

Classical parity checks use a binary matrix H_c and syndrome $s = H_c x$ over F_2. Error behavior belongs to the selected code and channel model. The shared architecture stores generators, syndromes, graph relations and their entity/address metadata.

> **Exact bridge**
>
> The original XOR symmetrization creates graph adjacency. Graph adjacency creates commuting Pauli generators. This construction supplies the quantum connection through explicit operators and a defined state space.

BOOLEAN FUNCTIONS / CODING / 22

Nonlinear Boolean and spectral channels

The existing four-term XOR is a linear map over F_2. It supplies fast parity and symmetric graph construction. A separate nonlinear function gives the source's proposed Boolean-spectrum connection an explicit realization.

$$g(x,y) = x \cdot y \pmod 2, \qquad x,y \in \mathbb{F}_2^k \tag{20.1}$$

$$W_g(a,b) = \sum_{x,y} (-1)^{g(x,y) + a\cdot x + b\cdot y} = 2^k (-1)^{a\cdot b} \tag{20.2}$$

The Walsh magnitude is exactly 2^k at every spectral coordinate. This is a bent function on 2k variables [R16]. Its number of ones is 2^(2k−1)−2^(k−1), as follows from $W_g(0,0) = 2^k$.

A four-bit example

$$g(x_1,x_2,x_3,x_4) = x_1x_2 \oplus x_3x_4 \tag{20.3}$$

Across 16 inputs, this function has six ones and all 16 Walsh coefficients have magnitude four. The test record reproduces those values.

Typed composition

A Boolean module stores its function, input partition and output interpretation. Its outputs can select geometry leaves, interaction edges or sampling patterns. Linear parity, nonlinear Boolean features and quantum stabilizers are three separately defined channels that can share packed storage.

A cryptographic construction defines its keyed state transformation. A quantum experiment defines preparation, operations and measurement. These application maps assign operational meaning to the formal functions.

> **Beyond the original mask**
>
> The architecture can carry both the original linear symmetrization and a maximally nonlinear Boolean channel. Their mathematical properties are explicit, so applications can select the operator appropriate to their domain.

RELATIVISTIC DOMAIN / 23

Log-time and 3+1 spacetime geometry

A spacetime adapter uses a three-dimensional spatial manifold Σ, for example a radial shell with angular coordinates (u,ψ). In this page, use geometrized units c = G = 1 [R17].

$$ds^2 = -N^2 dT^2 + h_{ij}(dx^i + \beta^i dT)(dx^j + \beta^j dT) \tag{21.1}$$

h is the spatial metric, N the lapse and β the shift. Their state is supplemented by extrinsic curvature K and matter energy/momentum fields.

$$K_{ij} = -\tfrac{1}{2N}(\partial_T h_{ij} - \mathcal{L}_\beta h_{ij}) \tag{21.2}$$

$$^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi\rho_E \tag{21.3}$$

$$D_j(K^{ij} - h^{ij}K) = 8\pi j^i \tag{21.4}$$

Here K = $h^{ij}K_ij$ is the trace, $\rho_E = T_{ab}\, n^a n^b$ the normal-frame energy density, and $j^i = -h^i{}_a\, n_b\, T^{ab}$ the projected momentum density. Equations (21.3)-(21.4) are the ADM Hamiltonian and momentum relations with the sign convention (21.2). Evolution equations and matter models supply the dynamics.

The log-time coordinate as an explicit lapse

$$t = t_*(e^T - 1), \qquad N_T = N_t\, t_* e^T, \qquad \beta^i_T = \beta^i_t\, t_* e^T \tag{21.5}$$

Time reparameterization multiplies the lapse and shift by dt/dT. For a static Euclidean shell with zero shift, $h_{rr} = 1$, $h_{uu} = r^2/(1-u^2)$, $h_{\psi\psi} = r^2(1-u^2)$ in its regular chart; K = 0. A unit t-lapse becomes $N_T = t_*e\hat{}T$.

This lifts the source’s log-time idea into a Lorentzian metric representation. Overlapping nonsingular angular charts or Cartesian coordinates cover the poles. Matter, curvature and field evolution occupy their own stored tensor fields.

SPACE / CYBER-PHYSICAL SYSTEMS / 24

Orbital geometry, networks and observation

An orbital entity stores ID, epoch, reference frame, position r and velocity v. The sphere index addresses direction; the radius and velocity remain physical state. Frame and ephemeris transformations follow a declared adapter [R18].

$$\mathbf{s} = \mathbf{r}/\|\mathbf{r}\|, \qquad w = \mathrm{locate}(R\mathbf{s}, d) \tag{22.1}$$

$$\ddot{\mathbf{r}} = -\frac{\mu_g}{\|\mathbf{r}\|^3}\mathbf{r} + \mathbf{a}_{\mathrm{control}} \tag{22.2}$$

For relative two-body position in nonrotating axes, $\mu_g = G(M+m)$, with GM as the test-mass specialization. Perturbations enter as additional acceleration. Coverage queries use spherical cells while the physical trajectory uses (22.2).

Communication geometry

$$\mathbf{y} = H\mathbf{x} + \mathbf{n}, \qquad \mathbf{x} = W\mathbf{s}_{\mathrm{signal}} \tag{22.3}$$

H is a complex channel matrix, W a beamforming map and n the channel noise. Antenna positions, wavelengths, delays and reference-frame transforms determine the chosen channel model. The hierarchy indexes endpoints and coverage regions; the network graph stores communication relations.

Observation and state estimation

$$\dot{x} = f(x,u), \qquad y = h(x) + \nu, \qquad \dot{\hat{x}} = f(\hat{x},u) + K(y - h(\hat{x})) \tag{22.4}$$

The same observation interface applies to robotics, instrumented materials and digital twins. Geometry supplies sensor locations or expected measurements; physical fields supply the evolving state; the observer maps measurements into estimated state.

An observation is timestamped with its frame and units. A controller acts through the corresponding physical port. This gives a concrete cross-layer connection among spatial lookup, field simulation, sensing and network control.

INTEGRATION / 25

Coupled profiles across the domains

A profile selects operators and connects typed outputs to typed inputs. It records its geometry program, material laws, fields, clock mapping, observation functions and evaluation schedule.

Profile	Explicit composition
Thermal geometric twin	CSG boundary → optical path integral → deposited heat Q_{abs} → temperature field → thermal deformation → updated geometry.
Buoyant plume	Volume cells → mass/momentum/energy fluxes → density/velocity/temperature → fitted cap/stem envelope → field rendering and observation.
Graph-neural geometry	Mask Q → graph features → learned residual field → constructive shape → samples → parameter-fitting update.
Replicated hierarchy	Tagged entity operations → set-union merge → deterministic records → organization and spatial lookups → display and aggregate queries.
Quantum graph authoring	Packed A/B inputs → symmetric Q → stabilizer tableau [I|Q] → graph-state preparation description.
Orbital network twin	Ephemeris/frame state → sphere direction/cell → coverage graph → channel model → measurement/control update.

An energy-coupled example

$$\dot{H}_{\text{mech}} = -P_{\text{diss}} + P_{\text{in}}, \qquad \dot{U}_{\text{thermal}} = P_{\text{diss}} - P_{\text{out}} \tag{23.1}$$

$$\frac{d}{dt}(H_{\text{mech}} + U_{\text{thermal}}) = P_{\text{in}} - P_{\text{out}} \tag{23.2}$$

The transfer term cancels when the subsystem balances are added. The same principle governs shared-face fluxes, mechanical/thermal coupling and power-conjugate control ports.

Direct execution

Events and continuous equations update the shared state through U. Empty and full-space geometry retain their defined occupancy values. Each coupling’s equations determine the evolution; source fields, boundary data and model parameters are part of the profile.

The expanded paradigm is therefore a composition of addressable structures, implicit geometry, continuous dynamics and discrete operators. Its domain connections are supplied by the explicit maps in this edition.

CHEMICAL / NON-EQUILIBRIUM DOMAINS / 26

Reaction fields and scale thermodynamics

A reaction-transport adapter uses molar concentrations c = (f,a,b,m)ᵀ in mol/m³, stoichiometric matrix N and rates v(c) in mol/(m³ s). Each reaction column records net species production [R19].

$$\partial_t c + \nabla \cdot (c\mathbf{v}_{\text{flow}}) = Nv(c) + \nabla \cdot (D\nabla c) + s_{\text{in}} - s_{\text{out}} \tag{24.1}$$

$$\ell^T N = 0 \quad \Longrightarrow \quad \ell^T N v(c) = 0 \tag{24.2}$$

A left-null vector l defines a conserved chemical amount. Boundary fluxes and supplied species determine its open-system change. Reaction enthalpies ΔH_r in J/mol contribute the thermal source −Σ_r ΔH_r v_r.

A boundary-producing catalytic example

$$F + B \longrightarrow A + B, \quad F + A \longrightarrow B + A, \quad F + A + B \longrightarrow M + A + B \tag{24.3}$$

$$v_1 = k_1 fb, \quad v_2 = k_2 fa, \quad v_3 = k_3 fab \tag{24.4}$$

F is a precursor, A and B are catalysts, and M is a boundary-forming product. These idealized species share a conserved monomer unit. The stoichiometric columns transfer one F unit to A, B or M, so l = (1,1,1,1) is conserved by the reactions. Rate constants carry the units appropriate to concentration and reaction order.

$$N_M = \int m \, dV, \quad V_M = \bar{v}_M N_M, \qquad R_o = \left(R_i^3 + \tfrac{3V_M}{4\pi}\right)^{1/3} \tag{24.4b}$$

Molar volume v̄_M converts the amount N_M into shell material volume V_M. R_i and R_o are its inner and outer radii. Boundary permeability and shape determine precursor influx; the internally generated product participates in regenerating the reaction domain [R20].

Scale-patterned transport and entropy

$$k(r) = k_0 e^{\varepsilon \sin(\omega \ln(r/r_*))}, \quad k_0 > 0, \; r_* > 0, \; r > 0 \tag{24.5a}$$

$$k(qr) = k(r), \quad q = e^{2\pi/|\omega|}, \quad \omega \neq 0 \tag{24.5b}$$

$$\dot{S}_{\text{gen, cond}} = \int_{\mathcal{D}} \frac{k\|\nabla\vartheta\|^2}{\vartheta^2} \, dV \geq 0, \quad \vartheta > 0 \tag{24.6}$$

ε and ω are dimensionless; ω = 0 gives constant conductivity. Scale-repeating conductivity gives the log-sine profile a material-transport role. Equation (24.6) is conductive entropy generation; reaction contributions enter separately [R21].

VALIDATION RECORD / I / 27

Equation checks: geometry and physics

The accompanying seeded calculation evaluates representative identities and update examples. It records the inputs and residuals for each check. Seed: 20261013; Python/NumPy double precision.

Check	Calculation	Result
Smooth CSG derivative	20,000 scalar pairs; central differences	Max residual 4.46×10^{-10}.
Smooth duality	20,000 pairs	$T_h(a,b) = -S_h(-a,-b)$.
Self-subtraction	20,000 values	$\max(a,-a)$ has empty negative interior.
Empty/full outcomes	Extended distances	Occupancies 0 and 1.
Equal-area profile	20,000 points, depth 14	Area residual $\le1.13\times10^{-17}$.
Cell representative radius	Same point sample	Max error 0.122318 rad; all within cell bounds.
Seam/pole indexing	Four designated points	Deterministic depth-14 addresses.
Parameter fitting	Sphere-radius objective, 12 steps	Loss 0.49 → 2.32×10^{-6}.
Finite-volume exchange	Three internal faces	Total amount 16 → 16.
Shell volumes	64 solid-angle cells	Sum equals $4\pi(3^3-2^3)/3$.
Drag power	10,000 relative velocities	Max residual 3.55×10^{-15}.
Closed drag/heat example	K=5.25 J initially	K_{next}=3.967865 J; heat=1.282135 J.
Coriolis work	10,000 vector pairs	Max absolute residual 2.00×10^{-15}.
Optical slab balance	I_{in}=7; κ=1.2; length=0.7	I_{out}=3.021974; absorbed=3.978026.
Port-Hamiltonian identity	1,000 states	Max energy-rate residual 8.88×10^{-16}.
Reaction stoichiometry	Three conserving reaction columns	Total-amount rate residual 9.71×10^{-17}.
Scale-patterned heat	10,000 radii; positive temperatures	Pattern residual 3.11×10^{-15}; conductive entropy ≥0.

The cell-radius row is an additional measurement within the equal-area-profile check. Random samples report observed residuals; exact cancellation, derivative and area identities are established by the equations on their respective pages.

VALIDATION RECORD / II / 28

Equation checks: information and charts

Check	Calculation	Result
Snapshot bit mutation	[0,1,1,0] XOR [1,0,1,0]	[1,1,0,0].
Rank/select topology	Five-node tree with a right-only node	Children and parents recovered from 10 shape bits.
Replica merge	All six orders of three operation sets	One final set; repeated delivery is idempotent.
Graph-state commutation	1,024 random 12-node graphs	$H\Lambda H^T = 0$ over F_2; rank 12.
Bent Boolean spectrum	All 16 inputs and spectral coordinates	Every Walsh magnitude is 4; six one-valued inputs.
Hyperbolic focus	1,000 point pairs	Max distance residual 4.00×10^{-15}.
Periodic phase	1,000 angles; a=0.35	Seam residual 1.78×10^{-15}; derivative ≥0.65.
Log-time chain rule	1,000 time coordinates	dt/dT = t+t_*.
Graph diffusion	Three-node path; signal [3,1,−2]	$x^TLx = 13$; total-signal derivative = 0.
Barycentric log-polar inverse	1,000 radii and angular codes	Radius and turn residuals below 10^{-13}.
Chamber radial inverse	1,000 radii; r_c=1, ell=0.8	Max residual below 10^{-12}.

Recorded result

All 27 scripted checks returned their expected result. They cover the geometry, physical-accounting, reaction, data-structure, distributed-algebra, quantum-graph and coordinate examples identified above.

Reproduction and model definition

The companion script *validate_xdtau_v13.py* writes *validation_results_v13.json*. The record includes the seed, sample counts, parameters and measured residuals. The edition's equations define the formal adapters; a chosen application profile supplies its material coefficients, observations, sources and discretization.

The integrated update can be reproduced from the encoded state, event sequence, program, model parameters and declared numerical scheme. View settings, physical time and random-generator state are recorded wherever they influence the selected profile.

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Edition lineage and attribution

Tom Klootwijk's XD-TAU Architectural Paradigm v1.1 and reviewed v1.2 supply the core. The 22-page diff force CSG Tom Klootwijk TAU architecture paradigm export supplies the expanded domain vocabulary. Version 1.3 formalizes those connections through the operators, fields and adapters printed here.

State evolution is direct. Empty and full-space geometry have explicit semantics. Mathematical domains specify where each representation and equation is defined. Paradigm attribution: Tom Klootwijk. Formal expansion, calculations and document preparation: AI-assisted.